

Homework 7 Solutions

Saturday, May 25, 2024 12:30 AM

5.4 #1

Proof. Let $f(x) = 1$. Then $F(\lambda) = \int_0^\infty e^{-\lambda x} dx$ which is absolutely integrable when $\operatorname{Re}(\lambda) > 0$ and not integrable when $\operatorname{Re}(\lambda) \leq 0$. When $\operatorname{Re}(\lambda) > 0$, $F(\lambda) = \int_0^\infty e^{-\lambda x} dx = -\frac{1}{\lambda} e^{-\lambda x} \Big|_0^\infty = \frac{1}{\lambda}$. Let $g(x) = x$. Then $G(\lambda) = \int_0^\infty x e^{-\lambda x} dx$ which is absolutely integrable when $\operatorname{Re}(\lambda) > 0$ and not integrable when $\operatorname{Re}(\lambda) \leq 0$. When $\operatorname{Re}(\lambda) > 0$, $G(\lambda) = \int_0^\infty x e^{-\lambda x} dx = x \left(-\frac{1}{\lambda} e^{-\lambda x}\right) \Big|_0^\infty - \int_0^\infty -\frac{1}{\lambda} e^{-\lambda x} dx = \frac{1}{\lambda^2}$. $\square$

5.4 #3

Proof. Note $F(\lambda) = \frac{\lambda}{(\lambda-2i)(\lambda+2i)}$ has simple poles at $\pm 2i$. So $f(x) = \operatorname{Res}(F(\lambda)e^{x\lambda}, 2i) + \operatorname{Res}(F(\lambda)e^{x\lambda}, -2i)$ which is $\frac{2i}{4i}e^{2ix} + \frac{-2i}{-4i}e^{-2ix} = \cos(2x)$. $\square$

5.4 #4

Proof. Note $F(\lambda) = \frac{1}{(\lambda-i)(\lambda+i)}$ has simple poles at $\pm i$. So $f(x) = \operatorname{Res}(F(\lambda)e^{x\lambda}, i) + \operatorname{Res}(F(\lambda)e^{x\lambda}, -i)$ which is $\frac{1}{2i}e^{ix} + \frac{1}{-2i}e^{-ix} = \sin(x)$. $\square$

6.1 #1

Proof. Identify $\mathbb{C} \cong \mathbb{R}^2$ by $z = x + iy = \begin{bmatrix} x \\ y \end{bmatrix}$. Complex multiplication, being $\mathbb{C}$ linear, is $\mathbb{R}$ -linear. Multiplication by 1 clearly corresponds to the identity matrix. Multiplication by i sends $x + iy$ to $-y + ix$, i.e., it takes $\begin{bmatrix} x \\ y \end{bmatrix}$ to $\begin{bmatrix} -y \\ x \end{bmatrix}$. So multiplication by i corresponds to $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Hence multiplication by $a + ib$ corresponds to $a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. In particular, multiplication by $e^{i\theta} = r \cos \theta + ir \sin \theta$ corresponds to $\begin{pmatrix} r \cos \theta & -r \sin \theta \\ r \sin \theta & r \cos \theta \end{pmatrix}$. $\square$

6.1 #2

Proof. The linear fractional transformation $f(z) = \frac{1+z}{1-z}$ is a conformal equivalence mapping the unit disk D to the right half plane $\operatorname{Re}(z) > 0$. Clearly $g(z) = z + 1$ is a conformal equivalence mapping $\operatorname{Re}(z) > 0$ to $\operatorname{Re}(z) > 1$ since $g' \equiv 1 \neq 0$ and g is 1-1. So $g \circ f$ is a conformal equivalence mapping D to $\operatorname{Re}(z) > 1$. The inverse $\frac{(z-1)-1}{(z-1)+1} = \frac{z-2}{z}$ is the desired map. $\square$

Section 6.1: 4

Solution. The map $w = i \frac{1+z}{1-z}$ is a conformal equivalence of the disc $\{|z| < 1\}$ to the upper half-plane $\{\operatorname{Im}(w) > 0\}$, and the map $w^{\frac{1}{4}}$ (principal branch) is a conformal equivalence of $\{\operatorname{Im}(w) > 0\}$ to the wedge $0 < \arg(z) < \pi/4$. So the desired map is

$$f(z) = \left(i \frac{1+z}{1-z} \right)^{\frac{1}{4}}.$$

Section 6.1: 5

Solution. By example 6.1.4, the map $w = \frac{1+z}{1-z}$ maps the unit disc 1-1 onto the set $\operatorname{Re}(w) > 0$. We write

$$\operatorname{Im}\left(\frac{1+z}{1-z}\right) = \operatorname{Im}\left(\frac{(1+z)(1-\bar{z})}{|1-z|^2}\right) = \frac{2 \operatorname{Im}(z)}{|1-z|^2}$$

So the upper disc $\operatorname{Im}(z) > 0$ goes to the first quadrant $\operatorname{Im}(w) > 0$, and the lower half $\operatorname{Im}(z) < 0$ to the 4th quadrant $\operatorname{Im}(w) < 0$.